

Problem Set 5

1. Butterworth magnitude response

For an $N = 4$ analog Butterworth lowpass filter,

$$|H_c(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}.$$

Compute $|H_c(j\Omega)|$ in dB at (i) $\Omega = \Omega_c$ and (ii) $\Omega = 2\Omega_c$.

2. Butterworth order and cutoff behavior

The Butterworth magnitude-squared response is

$$|H_c(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}.$$

(a) Derive the standard Butterworth low-pass order formula that guarantees a passband attenuation $\leq A_p$ in dB at Ω_p in rad/sec and a stopband attenuation $\geq A_s$ in dB at Ω_s in rad/sec:

$$N \geq \frac{\log_{10} \left[\frac{10^{A_s/10} - 1}{10^{A_p/10} - 1} \right]}{2 \log_{10} (\Omega_s/\Omega_p)}$$

Hints: Write $|H_c(j\Omega_p)|^2$ and $|H_c(j\Omega_s)|^2$ in terms of N , form a ratio, and solve for N .

(b) For $A_p = 1$ dB at $\Omega_p = 1$ rad/sec and $A_s = 40$ dB at $\Omega_s = 3$ rad/sec, compute the minimum required order N . Round up to the nearest integer and explain why you round up rather than down.

(c) If you change Ω_s in part (b) to $\Omega_s = 2$ rad/sec, a 33% change in Ω_s , how does N change?

3. Stability under the bilinear transform

The bilinear transform is

$$z = \frac{1 + sT/2}{1 - sT/2}.$$

(a) For $T = 0.1$ sec and analog pole

$$s_p = -2 + j3,$$

compute the mapped digital pole and verify that it lies inside the unit circle.

(b) Determine the warped digital frequency ω from the analog frequency Ω using

$$\omega = 2 \tan^{-1}(\Omega T/2)$$

and verify your result in (a).

4. Prewarping design exercise

You need a digital Butterworth low-pass filter with $T = 1$ (i.e., $f_s = 1$ Hz, normalized) and a digital passband edge at $\omega_p = 0.3\pi$ where the response should be at -1 dB, and a digital stopband edge at $\omega_s = 0.6\pi$ where the response should be at least -30 dB down.

(a) Prewarp both edges to analog frequencies Ω_p and Ω_s using

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right).$$

(b) Using your formula from problem 2, compute the required Butterworth order N .

(c) In Python, design the analog Butterworth prototype at the prewarped Ω_p (or Ω_c , your choice of convention) using `scipy.signal.butter(..., analog=True)`, then apply `scipy.signal.bilinear` (or `bilinear_zpk`) to obtain the digital filter. Plot the digital magnitude response and confirm the -1 dB and -30 dB points land at (approximately) 0.3π and 0.6π .

(d) Compute and plot the group delay $\tau_g(\omega)$ over $0 \leq \omega \leq \pi$ using `scipy.signal.group_delay`.

(e) Repeat the design in (c) without prewarping (i.e., using $\Omega_p = \omega_p$, $\Omega_s = \omega_s$ directly as analog frequencies) and show on the same plot how far the resulting digital cutoff misses the intended target. Quantify the error in normalized frequency.

5. Window method as frequency-domain convolution (ECEN 5632 Problem)

The window method result $H_w(e^{j\omega}) = \frac{1}{2\pi} H_d(e^{j\omega}) * W(e^{j\omega})$ is a periodic convolution.

(a) Explain intuitively why a *narrower* main lobe of $W(e^{j\omega})$ leads to a *sharper* transition band in $H_w(e^{j\omega})$.

(b) Explain intuitively why *lower* sidelobes in $W(e^{j\omega})$ lead to *lower* ripple in $H_w(e^{j\omega})$, and why these two goals (narrow main lobe, low sidelobes) are in tension for a fixed window length N . (Hint: think about the time-domain taper's smoothness versus its duration.)